

Nonlinear Dynamics, Phase-Locking, and Collective Synchronization in Spin Torque Oscillators

Student Number: s1116231
Faculty of Science, Radboud Universiteit
(Dated: June 1, 2026)

CONTENTS

I	Introduction	3
II	Methods	3
III	Analytical Analysis	3
A	Motivation of the LLGS Equation and Effective Field Components	3
B	Coordinate Transformation and Dimensional Reduction	3
C	Converting the LLGS Equation from Implicit to Explicit	4
D	Projecting the Explicit Form onto the $\{\mathbf{m}, \hat{\mathbf{e}}_\theta, \hat{\mathbf{e}}_\phi\}$ Basis	4
1	Projecting $\mathbf{H}_{\text{eff}}$ onto $\{\mathbf{m}, \hat{\mathbf{e}}_\theta, \hat{\mathbf{e}}_\phi\}$	4
2	Projecting $\mathbf{M}$ onto $\{\mathbf{m}, \hat{\mathbf{e}}_\theta, \hat{\mathbf{e}}_\phi\}$	4
E	Final Projection	4
IV	Numerical Results and Discussion	4
V	Conclusion	4
	References	5
A	Mathematical Derivations	6

I. INTRODUCTION

II. METHODS

III. ANALYTICAL ANALYSIS

A. Motivation of the LLGS Equation and Effective Field Components

The fundamental equation governing the magnetization dynamics of an STO is the Landau-Lifshitz-Gilbert-Slonczewski (LLGS) equation:

$$\frac{d\mathbf{m}}{dt} = -|\gamma| \mathbf{m} \times \mathbf{H}_{\text{eff}} + \alpha \mathbf{m} \times \frac{d\mathbf{m}}{dt} + |\gamma| \beta \mathbf{J} [\mathbf{m} \times (\mathbf{m} \times \mathbf{M})] \quad (1)$$

Here $\gamma = 1.9 \times 10^{11} \text{ Hz/T}$ is the gyromagnetic ratio, $\alpha = 10^{-2}$ is the Gilbert damping parameter and $\beta = 10/3$ represents material parameter and fundamental constants. $\mathbf{J}$ is the current flowing from the fixed to the free layer.

The first term in the LLGS equation is the larmor precession:

$$-|\gamma| \mathbf{m} \times \mathbf{H}_{\text{eff}} \quad (2)$$

This term describes how the magnetization rotates around a local effective field $\mathbf{H}_{\text{eff}}$. Because of the cross product ($\mathbf{m} \times \mathbf{H}_{\text{eff}}$), the resulting torque vector is pointing perpendicular to both the magnetization $\mathbf{m}$ and the field $\mathbf{H}_{\text{eff}}$. This term is conservative, it does not change the angle between $\mathbf{m}$ and $\mathbf{H}_{\text{eff}}$, meaning it causes a perpetual, circular rotation around the field vector without losing energy. The speed of this rotation is determined by $|\gamma|$.

The second term is the Gilbert damping described by:

$$\alpha \mathbf{m} \times \frac{d\mathbf{m}}{dt} \quad (3)$$

In reality, a magnetic system cannot precess forever as it loses energy to its environment. This energy relaxation is captured by the dimensionless Gilbert damping parameter α . This term takes the cross product of the magnetization with its own velocity ($\mathbf{m} \times \dot{\mathbf{m}}$) which creates a torque vector pointing directly inward towards the direction of $\mathbf{H}_{\text{eff}}$. This drags the magnetization vector down forcing it to spiral tightly until it aligns completely parallel with the lowest energy state along $\mathbf{H}_{\text{eff}}$.

The last term in the LLGS equation is the addition discovered by John Slonczewski in 1996 [1]:

$$|\gamma| \beta \mathbf{J} [\mathbf{m} \times (\mathbf{m} \times \mathbf{M})] \quad (4)$$

When an electrical current $\mathbf{J}$ passes through a fixed, thick magnetic layer $\mathbf{M}$, the electrons traveling through it become "spin-polarized"—meaning their quantum spins line up with $\mathbf{M}$. When these polarized electrons enter the dynamic free layer $\mathbf{m}$, they crash into its local magnetic moments and transfer their physical angular momentum to it. Depending on the direction of $\mathbf{J}$, this generated torque vector can point in the exact opposite direction on the Gilbert damping term and thus acts like an anti-damping mechanism.

An important factor in the LLGS equation is the effective magnetic field $\mathbf{H}_{\text{eff}}$ which consists of the external magnetic field $H_a = 0.2$ along the x-axis, an anisotropy field $H_k = 0.05$ also along the x-axis and a demagnetization field $H_{dz} = 4\pi/\mu_0 M_s = 1.6$ along the z-axis, where M_s is the saturation magnetization of the free layer. The effective magnetic field is expressed as:

$$\mathbf{H}_{\text{eff}} = H_a \hat{\mathbf{e}}_x + H_k m_x \hat{\mathbf{e}}_x - H_{dz} m_z \hat{\mathbf{e}}_z \quad (5)$$

B. Coordinate Transformation and Dimensional Reduction

To analytically solve the trajectory of the magnetization vector $\mathbf{m}$ the three-dimensional Cartesian system must be mapped onto a spherical coordinate system. Because $\mathbf{m}$ is a unit vector ($|\mathbf{m}| = 1$), the state space is restricted to the surface of a three-dimensional unit sphere. A sphere is a two-dimensional manifold, consequently the position of $\mathbf{m}$ can therefore be defined using only two independent angular degrees of freedom, the polar angle $0 \leq \theta \leq \pi$ and the azimuthal angle $0 \leq \phi \leq 2\pi$.

The exact algebraic transformation mapping the Cartesian components into the independent coordinate pair (θ, ϕ) is defined as:

$$m_x = \sin \theta \cos \phi \quad (6)$$

$$m_y = \sin \theta \sin \phi \quad (7)$$

$$m_z = \cos \theta \quad (8)$$

Through this geometric mapping, the dynamic problem is reduced from three coupled Cartesian differential equations down to a 2D planar phase space spanned entirely by θ and ϕ .

In order to describe the LLGS equation onto this 2D surface, we define a new orthonormal basis $\{\mathbf{m}, \hat{\mathbf{e}}_\theta, \hat{\mathbf{e}}_\phi\}$, the unit tangent vectors, fully derived in Equation A1 and Equation A2, are defined as:

$$\hat{\mathbf{e}}_\theta = \frac{\partial \mathbf{m} / \partial \theta}{|\partial \mathbf{m} / \partial \theta|} = \begin{pmatrix} \cos \theta \cos \phi \\ \cos \theta \sin \phi \\ -\sin \theta \end{pmatrix} \quad (9)$$

$$\hat{\mathbf{e}}_\phi = \frac{\partial \mathbf{m} / \partial \phi}{|\partial \mathbf{m} / \partial \phi|} = \begin{pmatrix} -\sin \phi \\ \cos \phi \\ 0 \end{pmatrix} \quad (10)$$

By applying the multivariate chain rule to the time-dependent state vector $\mathbf{m}(\theta(t), \phi(t))$, the total time derivative (velocity) of the magnetization is cleanly resolved into the local spherical basis as:

$$\frac{d\mathbf{m}}{dt} = \frac{d\theta}{dt} \hat{\mathbf{e}}_\theta + \frac{d\phi}{dt} \sin \theta \hat{\mathbf{e}}_\phi \quad (11)$$

where $\frac{d\theta}{dt}$ and $\frac{d\phi}{dt}$ represent the real-time scalar angular velocities, the full derivation can be found in Equation A3

C. Converting the LLGS Equation from Implicit to Explicit

In order to analyze the LLGS equation the implicit form, with $d\mathbf{m}/dt$ on both sides, has to be converted to an explicit notation, with $d\mathbf{m}/dt$ on one side. To convert this we first take $\mathbf{m} \times d\mathbf{m}/dt$, which is fully derived in Equation A4 and takes on the form:

$$\mathbf{m} \times \frac{d\mathbf{m}}{dt} = -|\gamma| \mathbf{m} \times (\mathbf{m} \times \mathbf{H}_{\text{eff}}) - \alpha \frac{d\mathbf{m}}{dt} - |\gamma| \beta \mathbf{J} (\mathbf{m} \times \mathbf{M}) \quad (12)$$

When this is substituted into the LLGS equation, it takes on the explicit form:

$$\begin{aligned} \frac{d\mathbf{m}}{dt} = & \frac{-|\gamma|}{1+\alpha^2} \mathbf{m} \times \mathbf{H}_{\text{eff}} - \frac{\alpha|\gamma|}{1+\alpha^2} \mathbf{m} \times (\mathbf{m} \times \mathbf{H}_{\text{eff}}) \\ & + \frac{|\gamma|\beta\mathbf{J}}{1+\alpha^2} \mathbf{m} \times (\mathbf{m} \times \mathbf{M}) - \frac{\alpha|\gamma|\beta\mathbf{J}}{1+\alpha^2} \mathbf{m} \times \mathbf{M} \end{aligned} \quad (13)$$

The full derivation can be found in Equation A5.

D. Projecting the Explicit Form onto the $\{\mathbf{m}, \hat{\mathbf{e}}_\theta, \hat{\mathbf{e}}_\phi\}$ Basis

To transform the explicit three-dimensional vector LLGS equation into a closed two-dimensional system of scalar differential equations, the vector fields must be projected onto the orthonormal basis $\{\mathbf{m}, \hat{\mathbf{e}}_\theta, \hat{\mathbf{e}}_\phi\}$. Using the dimensionless time variable $\tau = \frac{\gamma}{1+\alpha^2} t$ the explicit form can be denoted as:

$$\frac{d\mathbf{m}}{d\tau} = -\mathbf{m} \times \mathbf{H}_{\text{eff}} - \alpha \mathbf{m} \times (\mathbf{m} \times \mathbf{H}_{\text{eff}}) + \beta \mathbf{J} \mathbf{m} \times (\mathbf{m} \times \mathbf{M}) - \alpha \beta \mathbf{J} \mathbf{m} \times \mathbf{M} \quad (14)$$

1. Projecting $\mathbf{H}_{\text{eff}}$ onto $\{\mathbf{m}, \hat{\mathbf{e}}_\theta, \hat{\mathbf{e}}_\phi\}$

The effective field strength $\mathbf{H}_{\text{eff}}$ can be projected onto the basis $\{\mathbf{m}, \hat{\mathbf{e}}_\theta, \hat{\mathbf{e}}_\phi\}$ as:

$$\mathbf{H}_{\text{eff}} = H_{\mathbf{m}} \mathbf{m} + H_\theta \hat{\mathbf{e}}_\theta + H_\phi \hat{\mathbf{e}}_\phi \quad (15)$$

Where the field components derived in Equation A6 are:

$$\begin{aligned} H_{\mathbf{m}} = & H_a \sin(\theta) \cos(\phi) + H_k \sin^2(\theta) \cos^2(\phi) \\ & - H_{dz} \cos^2(\theta) \\ H_\theta = & H_a \cos(\theta) \cos(\phi) + H_k \sin(\theta) \cos(\theta) \cos^2(\phi) \\ & + H_{dz} \cos(\theta) \sin(\theta) \\ H_\phi = & -H_a \sin(\phi) - H_k \sin(\theta) \sin(\phi) \cos(\phi) \end{aligned} \quad (16)$$

Using the identities:

$$\begin{aligned} \mathbf{m} \times \mathbf{m} &= 0 \\ \mathbf{m} \times \hat{\mathbf{e}}_\theta &= \hat{\mathbf{e}}_\phi \\ \mathbf{m} \times \hat{\mathbf{e}}_\phi &= -\hat{\mathbf{e}}_\theta \end{aligned} \quad (17)$$

We can find:

$$\mathbf{m} \times \mathbf{H}_{\text{eff}} = -H_\phi \hat{\mathbf{e}}_\theta + H_\theta \hat{\mathbf{e}}_\phi \quad (18)$$

And:

$$\mathbf{m} \times (\mathbf{m} \times \mathbf{H}_{\text{eff}}) = -H_\theta \hat{\mathbf{e}}_\theta - H_\phi \hat{\mathbf{e}}_\phi \quad (19)$$

2. Projecting $\mathbf{M}$ onto $\{\mathbf{m}, \hat{\mathbf{e}}_\theta, \hat{\mathbf{e}}_\phi\}$

For the magnetization in the fixed layer we take $\mathbf{M} = -\hat{\mathbf{e}}_x$, which can be expressed in the polar basis as:

$$\mathbf{M} = -\sin(\theta) \cos(\phi) \mathbf{m} - \cos(\theta) \cos(\phi) \hat{\mathbf{e}}_\theta + \sin(\phi) \hat{\mathbf{e}}_\phi \quad (20)$$

Again using the identities in Equation 17 we can find:

$$\mathbf{m} \times \mathbf{M} = -\sin(\phi) \hat{\mathbf{e}}_\theta - \cos(\theta) \cos(\phi) \hat{\mathbf{e}}_\phi \quad (21)$$

And:

$$\mathbf{m} \times (\mathbf{m} \times \mathbf{M}) = \cos(\theta) \cos(\phi) \hat{\mathbf{e}}_\theta - \sin(\phi) \hat{\mathbf{e}}_\phi \quad (22)$$

E. Final Projection

Now all the components to project the full LLGS equation into the form of Equation 11 are ready, this results in:

$$\frac{d\mathbf{m}}{d\tau} = () \quad (23)$$

IV. NUMERICAL RESULTS AND DISCUSSION

V. CONCLUSION

-
- [1] J. Slonczewski, Journal of Magnetism and Magnetic Materials
159, L1 (1996).

Appendix A: Mathematical Derivations

$$\begin{aligned}
 \hat{\mathbf{e}}_\theta &= \frac{\frac{\partial \mathbf{m}}{\partial \theta}}{\left| \frac{\partial \mathbf{m}}{\partial \theta} \right|} = \frac{\begin{pmatrix} \frac{\partial}{\partial \theta} \sin(\theta) \cos(\phi) \\ \frac{\partial}{\partial \theta} \sin(\theta) \sin(\phi) \\ \frac{\partial}{\partial \theta} \cos(\theta) \end{pmatrix}}{\left| \begin{pmatrix} \frac{\partial}{\partial \theta} \sin(\theta) \cos(\phi) \\ \frac{\partial}{\partial \theta} \sin(\theta) \sin(\phi) \\ \frac{\partial}{\partial \theta} \cos(\theta) \end{pmatrix} \right|} = \frac{\begin{pmatrix} \cos(\theta) \cos(\phi) \\ \cos(\theta) \sin(\phi) \\ -\sin(\theta) \end{pmatrix}}{\left| \begin{pmatrix} \cos(\theta) \cos(\phi) \\ \cos(\theta) \sin(\phi) \\ -\sin(\theta) \end{pmatrix} \right|} = \frac{\begin{pmatrix} \cos(\theta) \cos(\phi) \\ \cos(\theta) \sin(\phi) \\ -\sin(\theta) \end{pmatrix}}{\sqrt{(\cos(\theta) \cos(\phi))^2 + (\cos(\theta) \sin(\phi))^2 + (-\sin(\theta))^2}} \\
 &= \frac{\begin{pmatrix} \cos(\theta) \cos(\phi) \\ \cos(\theta) \sin(\phi) \\ -\sin(\theta) \end{pmatrix}}{\sqrt{(\cos^2(\theta) (\cos^2(\phi) + \sin^2(\phi))) + \sin^2(\theta)}} = \frac{\begin{pmatrix} \cos(\theta) \cos(\phi) \\ \cos(\theta) \sin(\phi) \\ -\sin(\theta) \end{pmatrix}}{\sqrt{\cos^2(\theta) + \sin^2(\theta)}} = \begin{pmatrix} \cos(\theta) \cos(\phi) \\ \cos(\theta) \sin(\phi) \\ -\sin(\theta) \end{pmatrix}
 \end{aligned} \tag{A1}$$

$$\begin{aligned}
 \hat{\mathbf{e}}_\phi &= \frac{\frac{\partial \mathbf{m}}{\partial \phi}}{\left| \frac{\partial \mathbf{m}}{\partial \phi} \right|} = \frac{\begin{pmatrix} \frac{\partial}{\partial \phi} \sin(\theta) \cos(\phi) \\ \frac{\partial}{\partial \phi} \sin(\theta) \sin(\phi) \\ \frac{\partial}{\partial \phi} \cos(\theta) \end{pmatrix}}{\left| \begin{pmatrix} \frac{\partial}{\partial \phi} \sin(\theta) \cos(\phi) \\ \frac{\partial}{\partial \phi} \sin(\theta) \sin(\phi) \\ \frac{\partial}{\partial \phi} \cos(\theta) \end{pmatrix} \right|} = \frac{\begin{pmatrix} -\sin(\theta) \sin(\phi) \\ \sin(\theta) \cos(\phi) \\ 0 \end{pmatrix}}{\left| \begin{pmatrix} -\sin(\theta) \sin(\phi) \\ \sin(\theta) \cos(\phi) \\ 0 \end{pmatrix} \right|} = \frac{\begin{pmatrix} -\sin(\theta) \sin(\phi) \\ \sin(\theta) \cos(\phi) \\ 0 \end{pmatrix}}{\sqrt{(-\sin(\theta) \sin(\phi))^2 + (\sin(\theta) \cos(\phi))^2 + (0)^2}} \\
 &= \frac{\begin{pmatrix} -\sin(\theta) \sin(\phi) \\ \sin(\theta) \cos(\phi) \\ 0 \end{pmatrix}}{\sqrt{\sin^2(\theta) (\sin^2(\phi) + \cos^2(\phi))}} = \frac{\begin{pmatrix} -\sin(\theta) \sin(\phi) \\ \sin(\theta) \cos(\phi) \\ 0 \end{pmatrix}}{\sqrt{\sin^2(\theta)}} = \frac{1}{\sin(\theta)} \begin{pmatrix} -\sin(\theta) \sin(\phi) \\ \sin(\theta) \cos(\phi) \\ 0 \end{pmatrix} = \begin{pmatrix} -\sin(\phi) \\ \cos(\phi) \\ 0 \end{pmatrix}
 \end{aligned} \tag{A2}$$

$$\begin{aligned}
 \frac{d\mathbf{m}}{dt} &= \frac{\partial \mathbf{m}}{\partial \theta} \frac{d\theta}{dt} + \frac{\partial \mathbf{m}}{\partial \phi} \frac{d\phi}{dt} = \begin{pmatrix} \cos(\theta) \cos(\phi) \\ \cos(\theta) \sin(\phi) \\ -\sin(\theta) \end{pmatrix} \frac{d\theta}{dt} + \begin{pmatrix} -\sin(\theta) \sin(\phi) \\ \sin(\theta) \cos(\phi) \\ 0 \end{pmatrix} \frac{d\phi}{dt} = \hat{\mathbf{e}}_\theta \frac{d\theta}{dt} + \sin(\theta) \begin{pmatrix} -\sin(\phi) \\ \cos(\phi) \\ 0 \end{pmatrix} \frac{d\phi}{dt} \\
 &= \hat{\mathbf{e}}_\theta \frac{d\theta}{dt} + \sin(\theta) \hat{\mathbf{e}}_\phi \frac{d\phi}{dt} = \frac{d\theta}{dt} \hat{\mathbf{e}}_\theta + \frac{d\phi}{dt} \sin(\theta) \hat{\mathbf{e}}_\phi = \dot{\theta} \hat{\mathbf{e}}_\theta + \dot{\phi} \sin \theta \hat{\mathbf{e}}_\phi
 \end{aligned} \tag{A3}$$

$$\begin{aligned}
 \mathbf{m} \times \frac{d\mathbf{m}}{dt} &= \mathbf{m} \times (-|\gamma| \mathbf{m} \times \mathbf{H}_{\text{eff}}) + \mathbf{m} \times \left(\alpha \mathbf{m} \times \frac{d\mathbf{m}}{dt} \right) + \mathbf{m} \times (|\gamma| \beta \mathbf{J} [\mathbf{m} \times (\mathbf{m} \times \mathbf{M})]) \\
 &= -|\gamma| \mathbf{m} \times (\mathbf{m} \times \mathbf{H}_{\text{eff}}) + \alpha \mathbf{m} \times \left(\mathbf{m} \times \frac{d\mathbf{m}}{dt} \right) + |\gamma| \beta \mathbf{J} \mathbf{m} \times [\mathbf{m} \times (\mathbf{m} \times \mathbf{M})]
 \end{aligned} \tag{A4}$$

Using the identity: $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B} (\mathbf{A} \cdot \mathbf{C}) - \mathbf{C} (\mathbf{A} \cdot \mathbf{B})$

$$= -|\gamma| \mathbf{m} \times (\mathbf{m} \times \mathbf{H}_{\text{eff}}) - \alpha \frac{d\mathbf{m}}{dt} - |\gamma| \beta \mathbf{J} (\mathbf{m} \times \mathbf{M})$$

$$\begin{aligned}
\frac{d\mathbf{m}}{dt} &= -|\gamma| \mathbf{m} \times \mathbf{H}_{\text{eff}} + \alpha \left(-|\gamma| \mathbf{m} \times (\mathbf{m} \times \mathbf{H}_{\text{eff}}) - \alpha \frac{d\mathbf{m}}{dt} - |\gamma| \beta \mathbf{J} (\mathbf{m} \times \mathbf{M}) \right) + |\gamma| \beta \mathbf{J} [\mathbf{m} \times (\mathbf{m} \times \mathbf{M})] \\
&= -|\gamma| \mathbf{m} \times \mathbf{H}_{\text{eff}} - \alpha |\gamma| \mathbf{m} \times (\mathbf{m} \times \mathbf{H}_{\text{eff}}) - \alpha^2 \frac{d\mathbf{m}}{dt} + |\gamma| \beta \mathbf{J} \mathbf{m} \times (\mathbf{m} \times \mathbf{M}) - \alpha |\gamma| \beta \mathbf{J} \mathbf{m} \times (\mathbf{m} \times \mathbf{M}) \\
\frac{d\mathbf{m}}{dt} + \alpha^2 \frac{d\mathbf{m}}{dt} &= -|\gamma| \mathbf{m} \times \mathbf{H}_{\text{eff}} - \alpha |\gamma| \mathbf{m} \times (\mathbf{m} \times \mathbf{H}_{\text{eff}}) + |\gamma| \beta \mathbf{J} \mathbf{m} \times (\mathbf{m} \times \mathbf{M}) - \alpha |\gamma| \beta \mathbf{J} \mathbf{m} \times (\mathbf{m} \times \mathbf{M}) \\
(1 + \alpha^2) \frac{d\mathbf{m}}{dt} &= -|\gamma| \mathbf{m} \times \mathbf{H}_{\text{eff}} - \alpha |\gamma| \mathbf{m} \times (\mathbf{m} \times \mathbf{H}_{\text{eff}}) + |\gamma| \beta \mathbf{J} \mathbf{m} \times (\mathbf{m} \times \mathbf{M}) - \alpha |\gamma| \beta \mathbf{J} \mathbf{m} \times (\mathbf{m} \times \mathbf{M}) \\
\frac{d\mathbf{m}}{dt} &= \frac{-|\gamma|}{1 + \alpha^2} \mathbf{m} \times \mathbf{H}_{\text{eff}} - \frac{\alpha |\gamma|}{1 + \alpha^2} \mathbf{m} \times (\mathbf{m} \times \mathbf{H}_{\text{eff}}) + \frac{|\gamma| \beta \mathbf{J}}{1 + \alpha^2} \mathbf{m} \times (\mathbf{m} \times \mathbf{M}) - \frac{\alpha |\gamma| \beta \mathbf{J}}{1 + \alpha^2} \mathbf{m} \times \mathbf{M}
\end{aligned} \tag{A5}$$

$$\begin{aligned}
\mathbf{H}_{\mathbf{m}} &= \mathbf{H}_{\text{eff}} \cdot \mathbf{m} = (H_a \hat{\mathbf{e}}_x + H_k m_x \hat{\mathbf{e}}_x - H_{dz} m_z \hat{\mathbf{e}}_z) \cdot (\sin(\theta) \cos(\phi) \hat{\mathbf{e}}_x + \sin(\theta) \sin(\phi) \hat{\mathbf{e}}_y + \cos(\theta) \hat{\mathbf{e}}_z) \\
&= (H_a + H_k \sin(\theta) \cos(\phi)) \sin(\theta) \cos(\phi) - H_{dz} \cos^2(\theta) \\
&= H_a \sin(\theta) \cos(\phi) + H_k \sin^2(\theta) \cos^2(\phi) - H_{dz} \cos^2(\theta) \\
\mathbf{H}_{\theta} &= \mathbf{H}_{\text{eff}} \cdot \hat{\mathbf{e}}_{\theta} = (H_a \hat{\mathbf{e}}_x + H_k m_x \hat{\mathbf{e}}_x - H_{dz} m_z \hat{\mathbf{e}}_z) \cdot (\cos(\theta) \cos(\phi) \hat{\mathbf{e}}_x + \cos(\theta) \sin(\phi) \hat{\mathbf{e}}_y - \sin(\theta) \hat{\mathbf{e}}_z) \\
&= (H_a + H_k \sin(\theta) \cos(\phi)) \cos(\theta) \cos(\phi) + H_{dz} \sin(\theta) \cos(\theta) \\
&= H_a \cos(\theta) \cos(\phi) + H_k \sin(\theta) \cos(\theta) \cos^2(\phi) + H_{dz} \sin(\theta) \cos(\theta) \\
\mathbf{H}_{\phi} &= \mathbf{H}_{\text{eff}} \cdot \hat{\mathbf{e}}_{\phi} = (H_a \hat{\mathbf{e}}_x + H_k m_x \hat{\mathbf{e}}_x - H_{dz} m_z \hat{\mathbf{e}}_z) \cdot (-\sin(\phi) \hat{\mathbf{e}}_x + \cos(\phi) \hat{\mathbf{e}}_y) \\
&= (H_a + H_k \sin(\theta) \cos(\phi)) (-\sin(\phi)) \\
&= -H_a \sin(\phi) - H_k \sin(\theta) \sin(\phi) \cos(\phi)
\end{aligned} \tag{A6}$$

$$\begin{aligned}
\frac{d\mathbf{m}}{d\tau} &= -\mathbf{m} \times \mathbf{H}_{\text{eff}} - \alpha \mathbf{m} \times (\mathbf{m} \times \mathbf{H}_{\text{eff}}) + \beta \mathbf{J} \mathbf{m} \times (\mathbf{m} \times \mathbf{M}) - \alpha \beta \mathbf{J} \mathbf{m} \times \mathbf{M} \\
&= H_{\phi} \hat{\mathbf{e}}_{\theta} - H_{\theta} \hat{\mathbf{e}}_{\phi} + \alpha H_{\theta} \hat{\mathbf{e}}_{\theta} + \alpha H_{\phi} \hat{\mathbf{e}}_{\phi} - \beta \mathbf{J} \sin(\phi) \hat{\mathbf{e}}_{\phi} + \beta \mathbf{J} \cos(\theta) \cos(\phi) \hat{\mathbf{e}}_{\theta} + \alpha \beta \mathbf{J} \sin(\phi) \hat{\mathbf{e}}_{\theta} + \alpha \beta \mathbf{J} \cos(\theta) \cos(\phi) \hat{\mathbf{e}}_{\phi} \\
&= (H_{\phi} + \alpha H_{\theta} + \beta \mathbf{J} \cos(\theta) \cos(\phi) + \alpha \beta \mathbf{J} \sin(\phi)) \hat{\mathbf{e}}_{\theta} + (-H_{\theta} + \alpha H_{\phi} - \beta \mathbf{J} \sin(\phi) + \alpha \beta \mathbf{J} \cos(\theta) \cos(\phi)) \hat{\mathbf{e}}_{\phi} \\
&= (-H_a \sin(\phi) - H_k \sin(\theta) \sin(\phi) \cos(\phi) + \alpha H_a \cos(\theta) \cos(\phi) + \alpha H_k \sin(\theta) \cos(\theta) \cos^2(\phi) \\
&\quad + H_{dz} \cos(\theta) \sin(\theta) + \beta \mathbf{J} \cos(\theta) \cos(\phi) + \alpha \beta \mathbf{J} \sin(\phi)) \hat{\mathbf{e}}_{\theta} \\
&\quad + (-H_a \cos(\theta) \cos(\phi) - H_k \sin(\theta) \cos(\theta) \cos^2(\phi) - H_{dz} \cos(\theta) \sin(\theta) - \alpha H_a \sin(\phi) - \alpha H_k \sin(\theta) \sin(\phi) \cos(\phi) \\
&\quad - \beta \mathbf{J} \sin(\phi) + \alpha \beta \mathbf{J} \cos(\theta) \cos(\phi)) \hat{\mathbf{e}}_{\phi} \\
&= ((\alpha H_a + \beta \mathbf{J}) \cos(\theta) \cos(\phi) + \alpha (H_{dz} + H_k \cos^2(\phi))) \sin(\theta) \cos(\theta) \\
&\quad + (-H_a + \alpha \beta \mathbf{J}) \sin(\phi) - H_k \sin(\theta) \sin(\phi) \cos(\phi)) \hat{\mathbf{e}}_{\theta} \\
&\quad + (- (\alpha H_a + \beta \mathbf{J}) \sin(\phi) - (H_{dz} + H_k \cos^2(\phi)) \cos(\theta) \sin(\theta) \\
&\quad + (-H_a + \alpha \beta \mathbf{J}) \cos(\theta) \cos(\phi) - \alpha H_k \sin(\theta) \sin(\phi) \cos(\phi)) \hat{\mathbf{e}}_{\phi}
\end{aligned} \tag{A7}$$